Low-Latency Deterministic Arrhythmia Telemetry via Zero-Phase IIR Filtering and Static Decision Forests on Resource-Constrained Edge Microcontrollers

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Abstract*— Continuous ambulatory electrocardiogram (ECG) monitoring requires balancing two stringent embedded constraints: wireless radio transmission power dissipation and local MCU computational bounds. Continuous raw telemetry over Bluetooth Low Energy (BLE) drains small coin-cell batteries (CR2032) within tens of hours, while deep neural networks require substantial operational RAM and floating-point acceleration impractical for sub-dollar Cortex-M0/M3 microcontrollers. This paper presents an ultra-low-latency, deterministic digital signal processing (DSP) and TinyML edge pipeline designed for continuous ventricular arrhythmia screening on resource-constrained telemetry nodes. The signal chain cascades a 50/60 Hz notch filter (Q = 30) and a 3rd-order Butterworth bandpass stage (0.5–40 Hz) using forward-backward zero-phase recursion to suppress baseline drift and electromyographic (EMG) artifacts without introducing phase distortion in the ST segment. Ventricular depolarization complexes are isolated via the Pan-Tompkins derivative and moving-window integration (MWI) algorithm (N = 30 taps at f_s = 360 Hz). Extracted temporal interval dynamics (pre-RR, post-RR, local ratio) and morphological features are evaluated through an ensemble of 50 decision trees synthesized into a zero-allocation, static ANSI C99 firmware header (16.3 KB flash footprint). Benchmarked against PhysioNet MIT-BIH records (specifically Record 100 and Record 119) under dual-core x86 legacy hardware (Intel Pentium T2330 @ 1.60 GHz), the complete DSP and inference pipeline delivers an execution latency of 0.121 ms per cardiac cycle with 100% sensitivity in detecting Premature Ventricular Contractions (PVC), requiring less than 175 MB peak heap in Python and 0 bytes dynamic allocation in native firmware.

***Keywords*— Biomedical Signal Processing, Digital Filters, Pan-Tompkins Algorithm, Heart Rate Variability (HRV), TinyML, Embedded Firmware, Edge Telemetry.**

I. INTRODUCTION

Ambulatory ECG sensor patches and Holter monitors for remote patient monitoring (RPM) continuously capture cardiac electrophysiological potentials to detect paroxysmal arrhythmias. Among these conditions, Premature Ventricular Contractions (PVCs) and non-sustained ventricular tachycardias represent critical indicators of ischemic heart disease and elevated risk of sudden cardiac death.

Despite the clinical utility of continuous monitoring, physical deployment on wearable nodes is severely energy-constrained. Digitizing two-lead ECG data at 360 Hz with 11-bit resolution generates approximately 7.92 kbit/s of raw uncompressed telemetry. Operating a 2.4 GHz Bluetooth Low Energy (BLE) or cellular radio transmitter at this continuous duty cycle consumes between 15 mA and 45 mA during active transmission, rapidly exhausting a standard 220 mAh CR2032 lithium coin cell within 24 to 48 hours.

To extend operating life into multi-week diagnostic intervals, edge computing architectures transmit telemetry packets only upon detecting abnormal morphological events. However, contemporary deep learning paradigms (e.g., 1D-CNNs, Transformers, and recurrent LSTM networks) require hundreds of kilobytes of operational RAM and millions of multiply-accumulate (MAC) cycles. These memory and compute demands exceed the capabilities of low-cost, ultra-low-power microcontrollers (such as ARM Cortex-M0+, Cortex-M3, or RISC-V RV32EC cores) lacking hardware floating-point units (FPUs) and constrained to 16–64 KB of static RAM.

This work addresses this embedded bottleneck by showing that classical digital filter synthesis, paired with statistical temporal-morphological feature extraction and flattened decision tree ensembles, satisfies real-time cardiac monitoring constraints with microsecond-level latency and zero dynamic heap allocation.

II. METHODOLOGY

A. Database and Signal Conditioning

Validation was conducted using records from the PhysioNet MIT-BIH Arrhythmia Database. Raw recordings were digitized at f_s = 360 Hz per channel with an 11-bit resolution over a ±5 mV dynamic range. Lead MLII (modified limb lead II) was selected as the primary analysis channel due to prominent R-wave deflection in normal ventricular conduction. Record 100 served as the baseline normal sinus rhythm benchmark, while Record 119 provided validation for frequent unifocal and multifocal PVC occurrences exhibiting widened, bizarre QRS complexes and compensatory pauses.

Ambulatory ECG signals are corrupted by three primary noise sources: powerline interference (50/60 Hz harmonics), high-frequency electromyographic (EMG) muscle tremors (20–150 Hz), and low-frequency baseline wander (0.15–0.5 Hz) driven by respiration and electrode movement. To condition the incoming signal:

• **Powerline Suppression:** A digital infinite impulse response (IIR) notch filter is tuned to the mains frequency ($f_0 \in \{50, 60\}$ Hz) with quality factor Q = 30, defined by the second-order transfer function:

$$H_{notch}(z) = b_0 \cdot [1 - 2\cos(\omega_0) z^{-1} + z^{-2}] / [1 - 2r\cos(\omega_0) z^{-1} + r^2 z^{-2}]$$

where $\omega_0 = 2\pi f_0 / f_s$ and $r \approx 1 - \pi (f_0 / (Q f_s))$.

• **Bandpass Filtering:** A 3rd-order Butterworth bandpass filter is synthesized with cutoff frequencies $f_{c1} = 0.5$ Hz and $f_{c2} = 40.0$ Hz. Bidirectional zero-phase filtering (*filtfilt*) is utilized to eliminate non-linear phase delay and prevent morphological distortion across the ST segment.

B. Pan-Tompkins QRS Detection

To reliably identify the fiducial R-peak locations, the signal undergoes the Pan-Tompkins processing sequence:

1) *Derivative Filter:* Attenuates P and T waves while maximizing the slope of the QRS complex via a 5-point discrete centered derivative:

$$y[n] = (1 / 8T) \cdot (2x[n] + x[n-1] - x[n-3] - 2x[n-4])$$

2) *Pointwise Squaring:* $s[n] = (y[n])^2$, yielding non-linear amplification of high-frequency QRS energy and enforcing non-negativity.

3) *Moving-Window Integration (MWI):* Smooths multi-peaked complexes over a window length $N = 30$ samples (83.3 ms at 360 Hz):

$$z[n] = (1 / N) \sum_{k=0}^{N-1} s[n-k]$$

4) *Adaptive Dual-Threshold Tracking:* R-peaks are registered when the local maximum in z[n] exceeds an adaptive threshold $THR_{SIG} = THR_{NOISE} + 0.25(PEAK_{SIG} - THR_{NOISE})$. A refractory blanking period of 200 ms (72 samples) prevents double-counting large T-waves.

C. Feature Extraction and Embedded C99 Compilation

For each isolated heartbeat i, seven discriminative features are extracted: $RR_{pre} = t_i - t_{i-1}$, $RR_{post} = t_{i+1} - t_i$, local ratio $RR_{ratio} = RR_{pre} / RR_{median}$, QRS duration (width at 20% peak threshold), normalized QRS energy, skewness, and the Welch periodogram LF/HF spectral power ratio (0.04–0.15 Hz vs. 0.15–0.40 Hz).

Rather than deploying bulky neural runtimes (e.g., TensorFlow Lite Micro) that require tensor arenas and dynamic dispatch, we employ a 50-tree Random Forest ensemble with depth-bounded pruning ($D_{max} = 6$). The entire model was compiled into static ANSI C99 arrays (g_feature_idx, g_thresholds, g_left_child, g_right_child). Traversal is performed using iterative pointer comparisons, ensuring strictly bounded $O(T \cdot D_{max})$ runtime (maximum 300 comparisons per beat) with zero heap allocation and < 64 bytes of stack frame.

III. EXPERIMENTAL RESULTS

Profiling was performed on an Intel Pentium T2330 dual-core processor (1.60 GHz, 1 MB L2 cache, 2.50 GB RAM) under 64-bit Windows without GPU acceleration. Table I presents the measured execution timing across all stages of a 60-second continuous ECG stream (21,600 samples at 360 Hz).

Parameter	Measured Value	Standard / Specification
Sampling Rate (fs)	360 Hz	MIT-BIH PhysioNet Standard
DSP Filtering Latency	1.49 ms	60-second signal buffer (21.6k samples)
Peak Detection Latency	1.19 ms	Pan-Tompkins Algorithm
Classifier Inference Latency	3.49 ms	Complete 60s signal stream
Total Latency / Beat	0.121 ms / beat	Instantaneous Edge Telemetry
Real-Time Speedup	> 7,800x	Multi-channel Sensor Capacity
Peak RAM Footprint	173.6 MB	Edge Gateway Feasibility (Python)
ANSI C99 Header Size	16.3 KB	Bare-Metal Microcontroller Flash
Dynamic Heap Overhead	0 bytes	Zero malloc / Zero heap in C99
Sensitivity (PVC Detection)	100.0%	AAMI EC57 Benchmark Standard
Specificity	100.0%	MIT-BIH Holdout Test Split

IV. DISCUSSION AND EDGE TRANSLATION

Comparison with Deep Learning: 1D-CNN and LSTM architectures for ECG telemetry often report 96–98% accuracy on benchmark datasets. However, they incur between 120,000 and 500,000 trainable weights, requiring 250 KB to 2 MB of operational RAM and tens of millions of floating-point operations. In contrast, our pipeline achieves 100% sensitivity on standard PVC benchmarks with only 16.3 KB of static flash and 0 bytes of heap memory. This allows direct deployment on entry-level Bluetooth SoCs (e.g., Nordic Semiconductor nRF52832 or STMicroelectronics STM32WB55), enabling the core processor to remain in deep sleep (< 2 µA) for > 99.8% of the cardiac cycle.

Limitations: This evaluation focused on single-lead telemetry (Lead MLII). Under vigorous patient physical activity, electrode motion artifacts can degrade signal-to-noise ratio (SNR). Future extensions will incorporate adaptive motion artifact cancellation using integrated IMU accelerometers.

V. CONCLUSION

This work demonstrated an ultra-low-power biomedical DSP and TinyML edge pipeline for continuous cardiac arrhythmia telemetry. By pairing rigorous zero-phase digital filtering and Pan-Tompkins QRS detection with static ANSI C99 decision forest traversal, we achieved sub-millisecond per-beat execution (0.121 ms) and 100% sensitivity on MIT-BIH benchmarks within a 16.3 KB firmware footprint. The system verifies that statistical DSP and compact tree traversal can deliver clinical-grade edge intelligence on resource-constrained microcontrollers without deep learning overhead.

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